

Qatar-1b

R-1701

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_161008 · created 2026-08-02T14:14:34+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2457669.74935 +/- 0.00088 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.192 +/- 0.056
Transit depth (Rp/Rs)^2	3.68 +/- 2.15 [%]
Orbital Inclination (inc)	88.81 +/- 2.86
Airmass coefficient 1 (a1)	0.785 +/- 0.087
Airmass coefficient 2 (a2)	0.13 +/- 0.1
Scatter in the residuals of the lightcurve fit is	11.31 %
Best Comparison Star	#2 - [167, 94]
Optimal Aperture	2.74
Optimal Annulus	7.08
Transit Duration (day)	0.0112 +/- 0.0033

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.192 $\pm$ 0.056 vs 0.1463 (deviation 0.73 σ)
Duration fit vs published	0.0112 d vs 0.0692 d (deviation 15.68 σ)
Mid-time O-C	2.2 min
Depth SNR	3.1
Residual scatter	11.31 %

Light curve

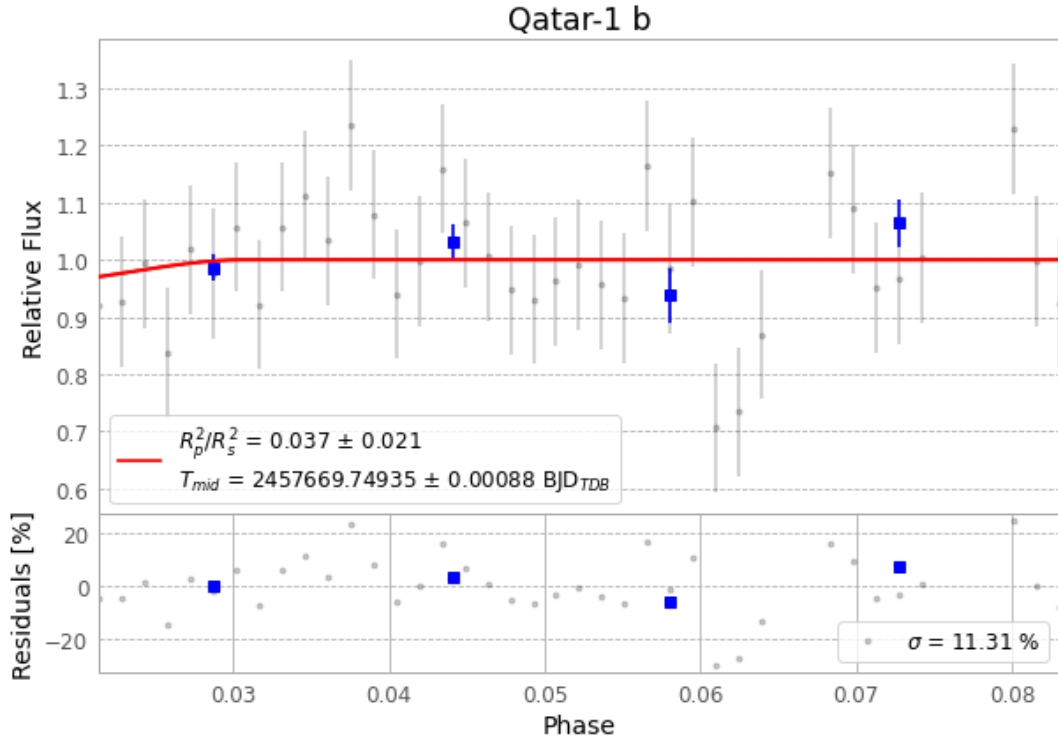


Figure 1 — FinalLightCurve_Qatar-1 b_2016-10-07.png (SHA-256 470c2388ba1c2bd8...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [191, 247], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 dd9bcf8015dee29c...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

43 FITS frames (28 MB), Qatar-1161008063912.FITS → Qatar-1161008084512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-161008.log (cf770b413256...), FinalLightCurve_Qatar-1 b_2016-10-07.png (470c2388ba1c...), FinalLightCurve_Qatar-1 b_2016-10-07.csv (796dfac41c2c...)

Compute resources

Wall time 300.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 14:14Z.